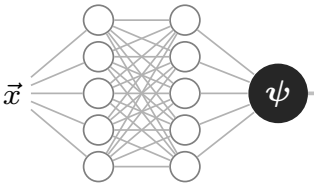
RUTGERS

Physics-Informed Neural Networks for Galactic Dark Matter

PINNs for Gaia kinematics and the diffuse γ-ray background

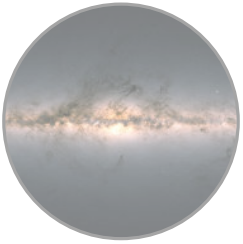
Eric Putney
Rutgers University, NHETC

Roadmap



Physics-informed neural networks

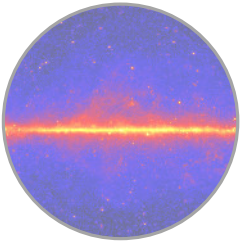
The PDE residual as the loss function



Part I: ClearPotential

Dark matter from the kinematics of 5.8 M *Gaia* stars

with M. Buckley, S. H. Lim, D. Shih



Part II: PINNCR

Diffuse γ-ray background to the GeV excess

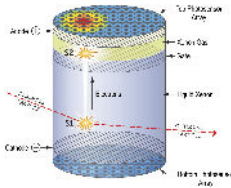
with Y. Park, D. Shih, M. Buckley, T. Slatyer, C. Chen

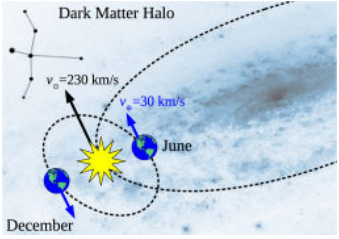
Images: ESA/Gaia/DPAC; NASA/DOE/Fermi-LAT Collaboration

The Milky Way as a dark matter laboratory

If DM couples to the SM, several channels:

- **Direct** detection (XENONnT, LZ): rate $\propto$ the local DM density $\rho_{\mathrm{DM},\odot}$





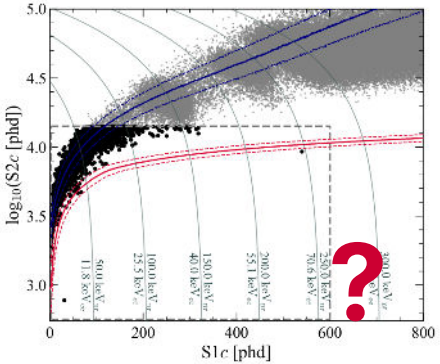
A xenon time-projection chamber, and the halo wind it looks for.

LZ Collaboration (Aalbers+22); Schumann 2019

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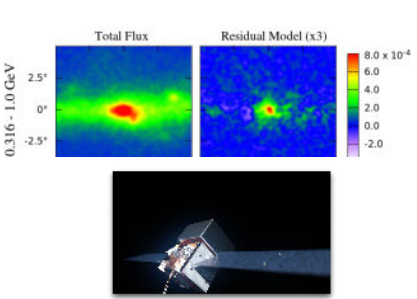
LZ science sample; the one event sits in the dashed box.

LZ Collaboration, arXiv:2609.02823, Fig. 4

The Milky Way as a dark matter laboratory

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- **Indirect** detection (Fermi-LAT): the Galactic Center GeV excess, over a large astrophysical background



The GeV excess and a DM-annihilation template; Fermi-LAT.

Daylan et al. 2014; NASA

The Milky Way as a dark matter laboratory

If DM couples to the SM, several channels:

- **Direct** detection (XENONnT, LZ): rate $\propto$ the local DM density $\rho_{\mathrm{DM},\odot}$
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- **Indirect** detection (Fermi-LAT): the Galactic Center GeV excess, over a large astrophysical background

Either way, the constraints depend on the Galactic distribution of DM, known only through gravity.

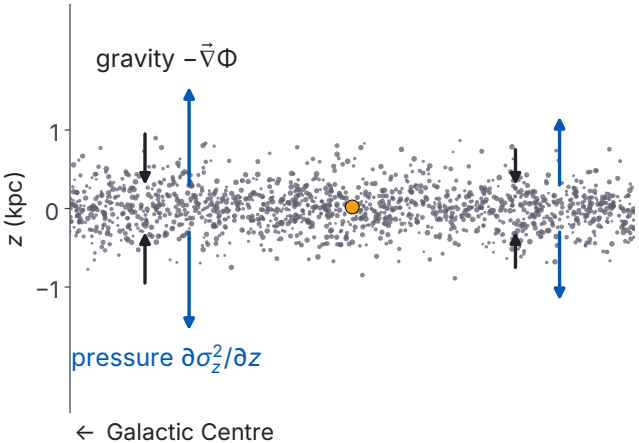


ESA/Gaia/DPAC, S. Payne-Wardenaar

The Oort problem

Classic dynamics problem: Oort estimated the local DM density in *1932*

- Stellar disk $\approx$ equilibrated gas
- Pressure ($\partial\sigma_z^2/\partial z$) pushes out, gravity ($-\vec{\nabla}\Phi$) pulls in



Vertical balance: the measured pressure out, gravity in; with DM, more pull to match it. Sun in gold.

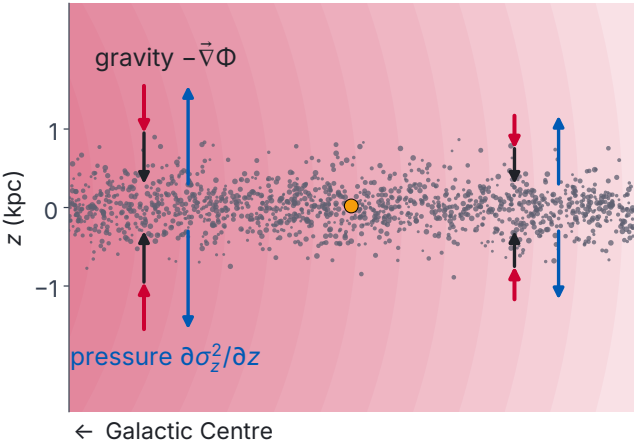
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Classic dynamics problem: Oort estimated the local DM density in *1932*

- Stellar disk $\approx$ equilibrated gas
- Pressure ($\partial\sigma_z^2/\partial z$) pushes out, gravity ($-\vec{\nabla}\Phi$) pulls in
- Stars trace the gravitational potential of **all** mass: stars, gas, DM

Vertical Jeans equation:

- moments of f, assumes symmetries
- not the general EoM



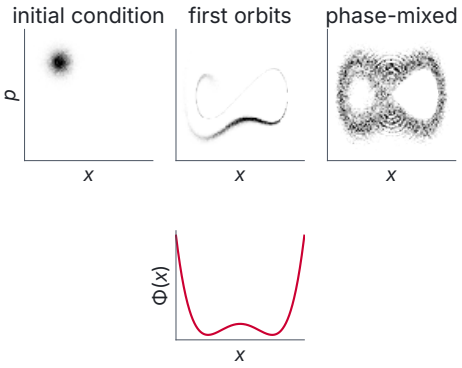
Vertical balance: the measured pressure out, gravity in; with DM, more pull to match it. Sun in gold.

The collisionless Boltzmann equation

Stars as a collisionless fluid in 6D phase space, $f(\vec{x}, \vec{v})$:

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \vec{\nabla}_x f - \vec{\nabla}_x \Phi \cdot \vec{\nabla}_v f = 0$$

- f fixes Φ; Poisson then gives ρ



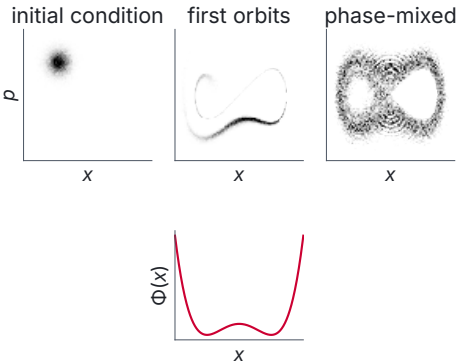
A blob of stars in a toy double well phase-mixes into an equilibrium f.

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- Equilibrium ($\partial_t f = 0$): orbits ~ 100 Myr $\ll$ 10 Gyr; one snapshot suffices
- Not perfectly relaxed (LMC, bar, spirals): assume it, test afterwards (backup)



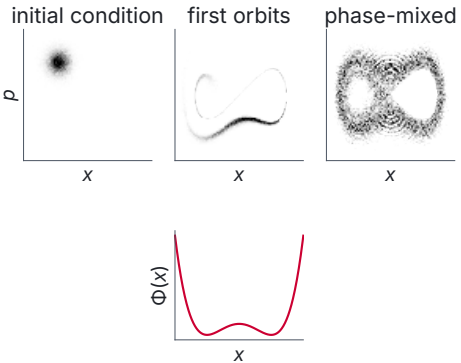
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- f fixes Φ; Poisson then gives ρ
- Equilibrium ($\partial_t f = 0$): orbits ~ 100 Myr $\ll$ 10 Gyr; one snapshot suffices
- Not perfectly relaxed (LMC, bar, spirals): assume it, test afterwards (backup)
- Needs f and its derivatives in 6D: millions of stars, a flexible model of f

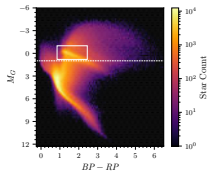


A blob of stars in a toy double well phase-mixes into an equilibrium f.

Measuring f: *Gaia* DR3

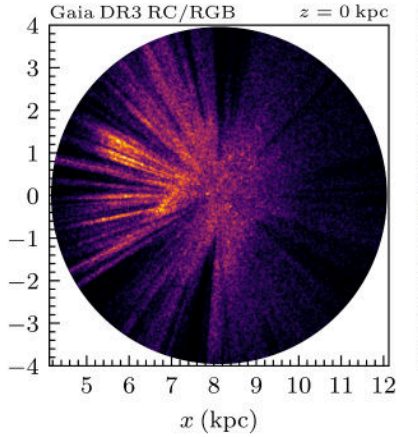
Gaia (2014–2025) mapped $\sim$2 B stars; DR3 (2022) uses the first 34 months:

- 33.8 M stars with full 6D phase space (1.46 B with 5D astrometry)
- Red clump + RGB tracers: 5.8 M within 4 kpc



$G_{\rm rvs}$ magnitude cut.

Lim et al. 2023



Tracers in the Galactic plane, Sun at center.

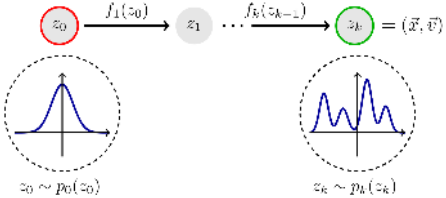
Putney et al. 2024

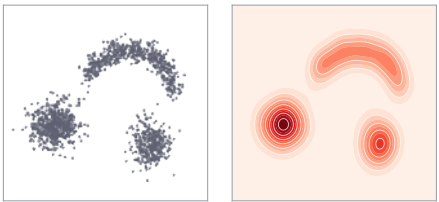
Normalizing flows

Normalizing flow: generative density estimation:

$$\vec{h} \to \vec{u} = g(\vec{h}), \quad p(\vec{u}) = p(g^{-1}(\vec{u})) \left| \det \frac{dg}{d\vec{h}} \right|^{-1}$$

- Invertible map g, latent $\vec{h} \to$ data $\vec{u}$, trained by maximum likelihood
- Smooth, differentiable $p(\vec{u})$: $\vec{\nabla}_x \ln f$, $\vec{\nabla}_v \ln f$ by autograd

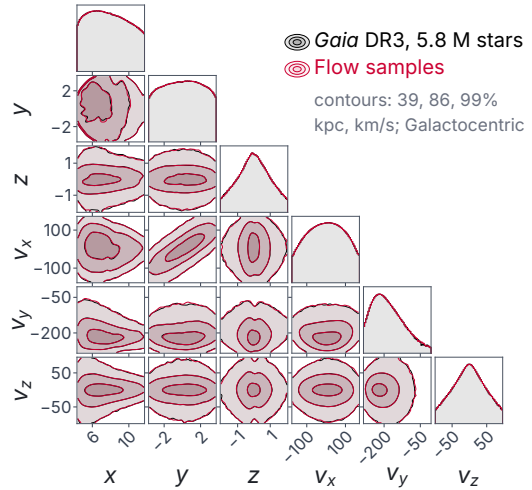




A 2D toy: points $\{\vec{u}_i\}$ and the flow's $p(\vec{u})$.

The flow fit to *Gaia* DR3

- $f = n(\vec{x})\, p(\vec{v}\,|\,\vec{x})$: two masked autoregressive flows, exact likelihood (backup)
- Reproduces the 6D marginals of 5.8 M stars



Gaia DR3 (black) vs. flow samples (red). *Lim et al. 2023*

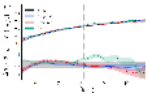
Solving the CBE for Φ

$\vec{a}(\vec{x})$ is independent of $\vec{v}$, and the flow gives any number of $\vec{v}$ at each $\vec{x}$:

$$\vec{v}_i \cdot \vec{\nabla}_x \ln f(\vec{x}, \vec{v}_i) - \vec{\nabla}_x \Phi \cdot \vec{\nabla}_v \ln f(\vec{x}, \vec{v}_i) = 0, \qquad i = 1, 2, \ldots$$

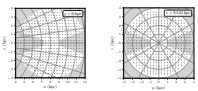
- Pointwise: least squares for $\vec{a}(\vec{x})$ at each $\vec{x}$ (N velocities, 3 unknowns) [Buckley+22, Lim+23]
- But Φ should be smooth and $\vec{a}$ curl-free: pointwise is expensive, incorrect

h277 hydro simulation: $\vec{a}$ to 1–5%



Buckley et al. 2022, arXiv:2205.01129

First application to data: *Gaia* DR3, 4 kpc



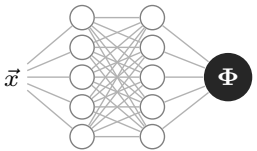
Lim et al. 2023, arXiv:2305.13358

Solving the CBE for Φ with a PINN

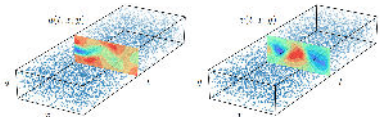
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- Pointwise: least squares for $\vec{a}(\vec{x})$ at each $\vec{x}$ (N velocities, 3 unknowns) [Buckley+22, Lim+23]
- But Φ should be smooth and $\vec{a}$ curl-free: pointwise is expensive, incorrect
- PINN: $\Phi_{\vartheta}(\vec{x})$ a neural network, the CBE residual is the loss, $\mathcal{L}_{\vartheta} = \left\langle |\mathrm{CBE}|^2 \right\rangle_{\vec{x},\vec{v}}$
- No grid: random points in the volume, held-out points to validate
- $\vec{a} = -\vec{\nabla}\Phi_{\vartheta}$ curl-free by construction



$\Phi_{\vartheta}(\vec{x})$ as a network; CBE residual is the loss.



Navier–Stokes. *Raissi et al., arXiv:1711.10566*

Learning Φ from a snapshot of phase space

Pairing a flow for f with a network for Φ on the CBE goes back to 2020:

2020 — **Deep Potential** Green & Ting: flow for f, network for Φ, CBE residual as loss

2021–22 — **Charting accelerations** An, Naik, Evans & Burrage: when Φ is unique

2022–23 — **Pointwise CBE** Buckley+22 on a simulation; Lim+23: first application to real data, *Gaia* DR3

2023–25 — **Deep Potential, *Gaia*** Green, Ting & Kamdar 23; Kalda & Green 25, with the pattern speed

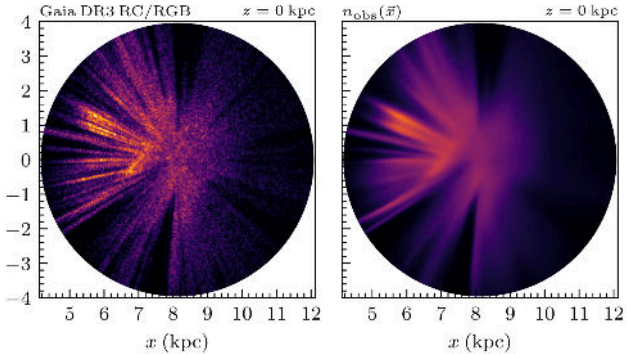
2024–25 — ClearPotential Φ-network again, plus dust (Putney+24, 25)

New here: dust ϵ learned inside the CBE · a 4 kpc volume · a full error covariance.

Dust removes stars

Optical surveys are obscured by dust lanes:

- Dims stars below *Gaia*'s radial-velocity limit ($G_{\rm RVS} \lesssim 14$)
- Missing stars bias f: a dust map cannot put them back



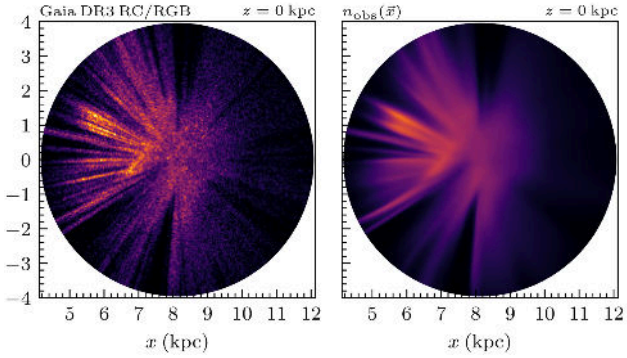
Gaia stars (left) vs. the flow's $n_{\rm obs}$ (right): the dust shadows are in f.

Putney et al. 2024

Dust removes stars

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- Dims stars below *Gaia*'s radial-velocity limit ($G_{\rm RVS} \lesssim 14$)
- Missing stars bias f: a dust map cannot put them back
- Worst in the midplane: derivatives of f amplify the shadows, biasing $\vec{a}$ and ρ



Gaia stars (left) vs. the flow's $n_{\rm obs}$ (right): the dust shadows are in f.

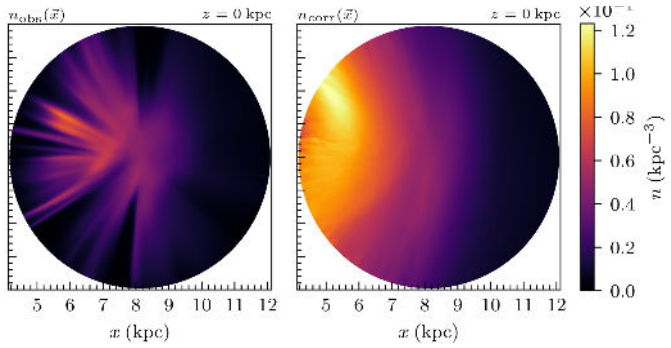
Putney et al. 2024

The dust-efficiency formalism

Relate the observed $f_{\rm obs}$ to the true f by a position-dependent efficiency $\epsilon(\vec{x})$:

$$f_{\rm obs}(\vec{x},\vec{v}) = \epsilon(\vec{x})\, f(\vec{x},\vec{v}) \qquad \Longrightarrow \qquad \vec{v}\cdot\vec{\nabla}\ln f_{\rm obs} - \vec{v}\cdot\vec{\nabla}\ln\epsilon - \vec{\nabla}\Phi\cdot\vec{\nabla}_v \ln f_{\rm obs} = 0$$

- **Second network** $\ln \epsilon_{\boldsymbol{\theta}}(\vec{x})$, same CBE loss



The flow's $n_{\rm obs}$ (left) and the dust-corrected $n_{\rm obs}/\epsilon$ (right) at $z = 0$.

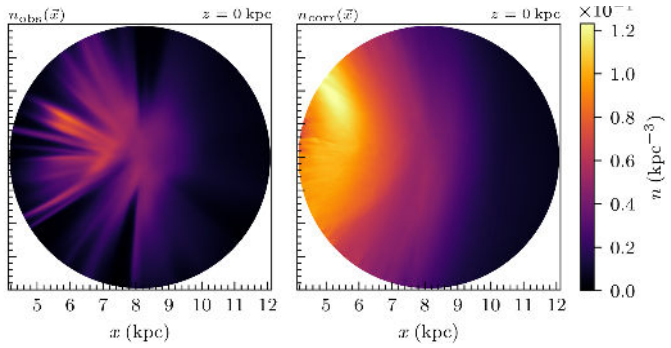
Putney et al. 2024

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- **Second network** $\ln \epsilon_{\boldsymbol{\theta}}(\vec{x})$, same CBE loss
- **Velocity-independent**: dust removes stars by position
- **Degeneracy broken**: $\vec{\nabla}\ln\epsilon$ couples to $\vec{v}$, $\vec{\nabla}\Phi$ to $\vec{\nabla}_v \ln f_{\rm obs}$



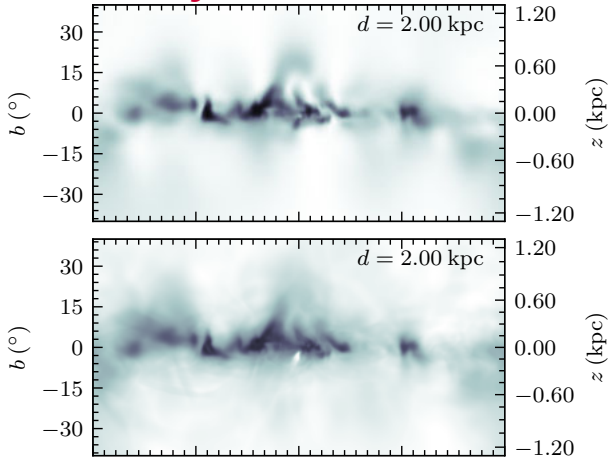
The flow's $n_{\rm obs}$ (left) and the dust-corrected $n_{\rm obs}/\epsilon$ (right) at $z = 0$.

Putney et al. 2024

Validating the learned dust efficiency

Independent check: ϵ_{L22} from the Lallement+22 3D dust map

- Red clump returns above the completeness cut



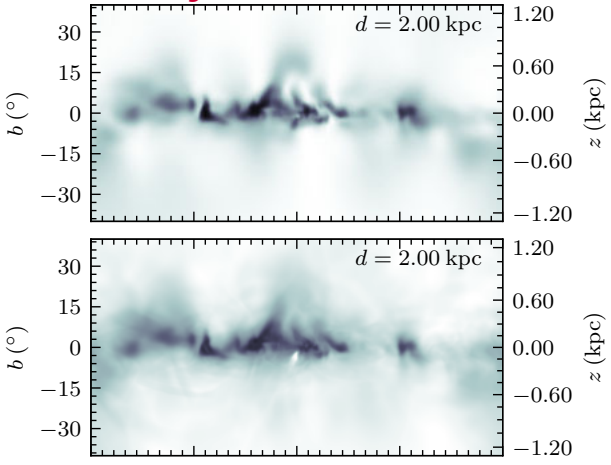
ϵ (top) vs. ϵ_{L22} (bottom) at $d = 2$ kpc.

Putney et al. 2024

Validating the learned dust efficiency

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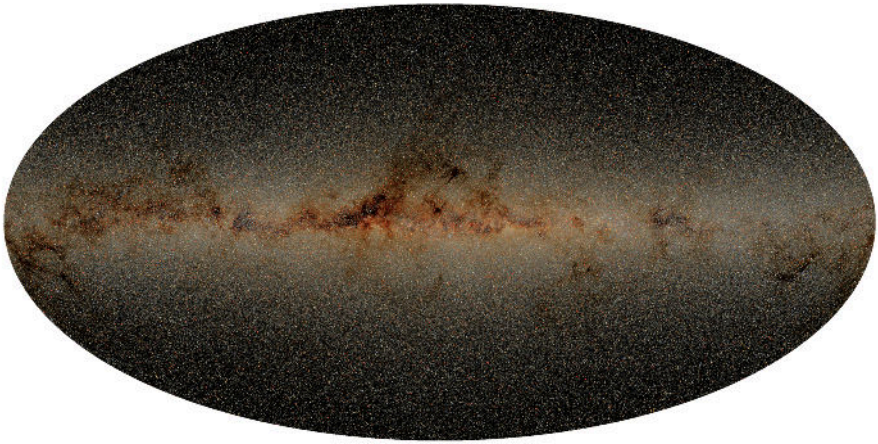
- Red clump returns above the completeness cut
- Same clouds, same shells at 1, 2, 3 kpc



ϵ (top) vs. $\epsilon_{\rm L22}$ (bottom) at $d = 2$ kpc.

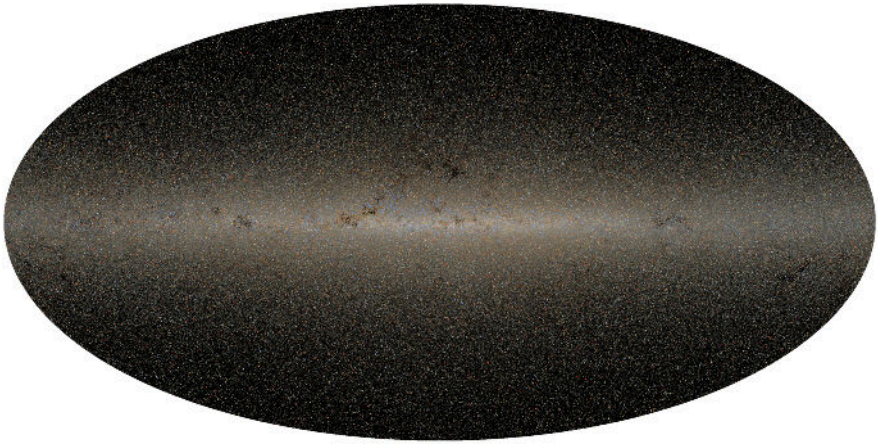
Putney et al. 2024

The dust-corrected Milky Way



Gaia DR3 tracers on the sky as observed: dust lanes carve the midplane. *Putney et al. 2024*

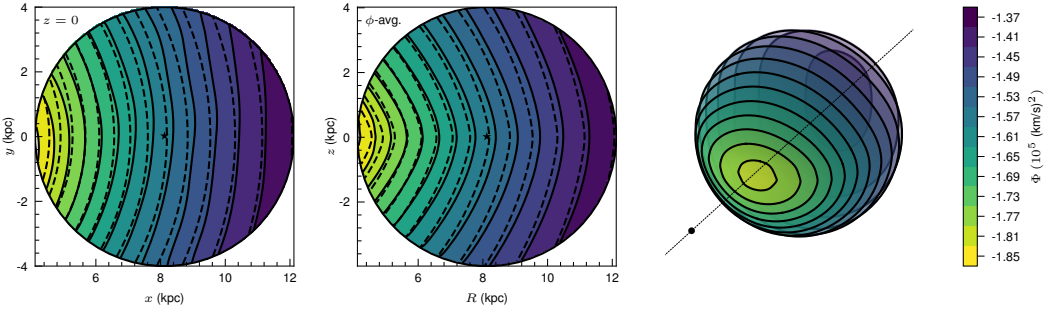
The dust-corrected Milky Way



The same stars, reweighted by $1/\epsilon$ and de-reddened. *Putney et al. 2024*

The potential and the acceleration field

$\Phi(\vec{x})$ over the 4 kpc volume, smooth and dust-corrected.

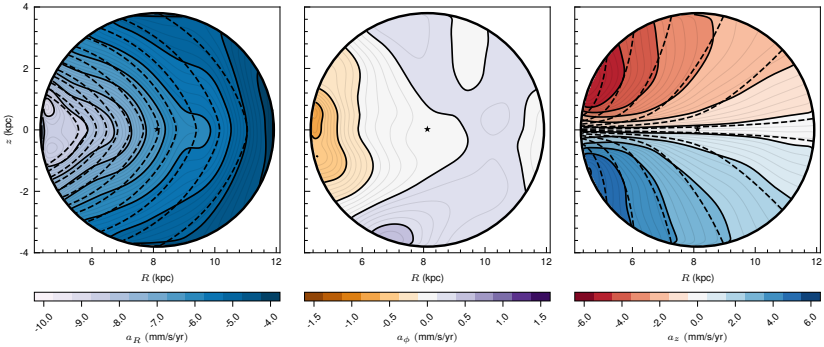


Φ: this work (solid) vs. MWPotential2014 (dashed): midplane, R–z, and isosurfaces around the Sun ($\star$).

Putney et al. 2025

The potential and the acceleration field

$\vec{a} = -\vec{\nabla}\Phi$: $a_\phi \neq 0$ at 3–5σ in the disk, so the field is not axisymmetric.



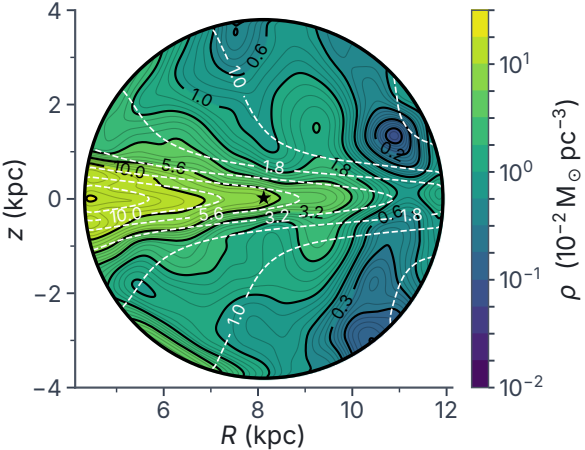
Left to right a_R, a_ϕ, a_z; dashed: MWPotential2014, which has $a_\phi \equiv 0$.

Putney et al. 2025

Mass density of the local Milky Way

$\rho = \nabla^2\Phi/4\pi G$ across the 4 kpc window:

- Stellar + gas disk at $|z| < 0.5$ kpc
- Non-zero density in the halo: DM
- Dip at $R \approx 11$ kpc, void above it



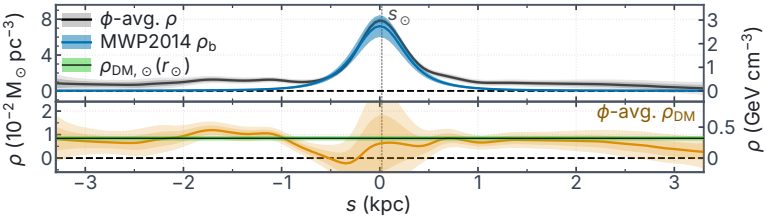
$\rho(R, z)$ (black) vs. MWPotential2014 (white dashed). *Putney et al. 2025*

The local dark matter density

- Subtract a baryonic model (stars + gas), average over the sphere $r = r_\odot$
- Errors: *Gaia*, statistical, training; systematics: the baryonic model

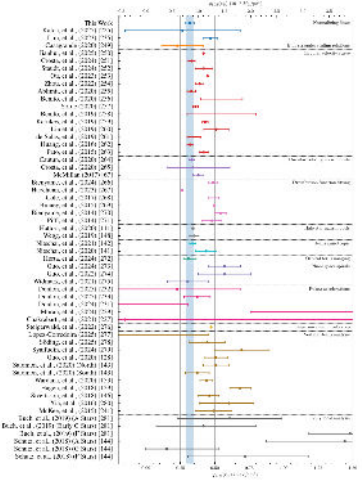
$$\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$$

spherical average at the Solar radius

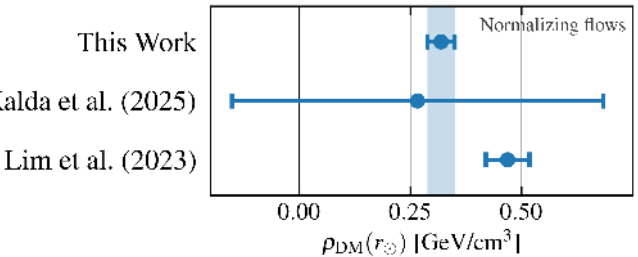


Total ρ and baryonic $\rho_{\rm b}$ (top); DM residual (bottom), along the arc through the Sun. *Putney et al. 2025*

The local dark matter density in context



One decade+ of $\rho_{\mathrm{DM},\odot}$.

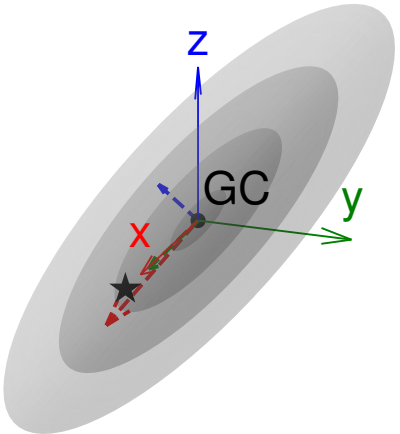


Putney et al. 2025

Halo shape

Tilted triaxial NFW fit:

- Axis ratios $b/a = 0.89 \pm 0.09$, $c/a = 0.33 \pm 0.06$: strongly oblate
- In-plane $\rho_{\mathrm{DM},\odot} = 0.43 \pm 0.03\,\mathrm{GeV/cm^3}$, 30% above the spherical average

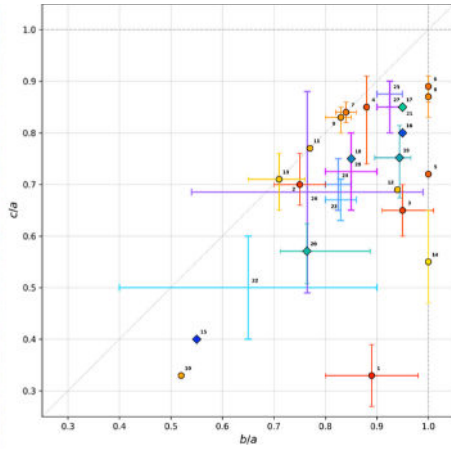


Best-fit tilted oblate halo. *Putney et al. 2025*

Halo shape

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- Dynamical fits: $c/a \approx 0.7$–0.9, ours is 0.33 (point 1); MW-like simulations (squares) span the same range

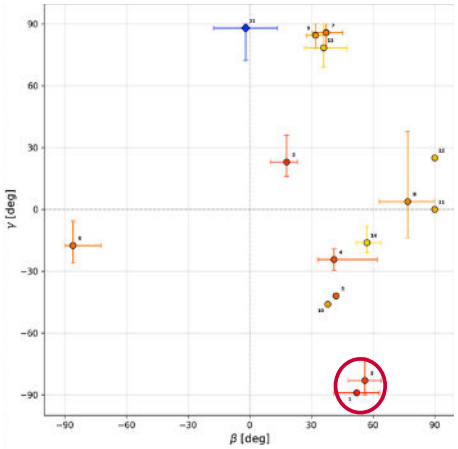


Preliminary. M. Scuderi, UNDARK 2 (Sept. 2026); Scuderi, Manconi & Cirelli, to appear.

Halo shape

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- Dynamical fits: $c/a \approx 0.7$–0.9, ours is 0.33 (point 1); MW-like simulations (squares) span the same range
- Orientation agrees with a GD-1 stream fit [Nibauer+25] (points 1 and 3)

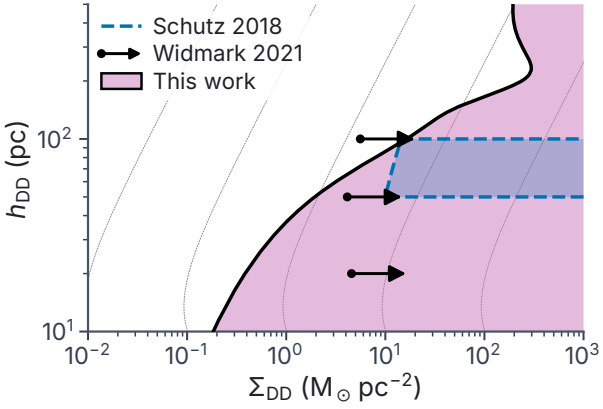


Preliminary. M. Scuderi, UNDARK 2 (Sept. 2026); Scuderi, Manconi & Cirelli, to appear.

A dark disk?

Dissipative dark sector, a disk aligned with the baryons [Fan+13]:

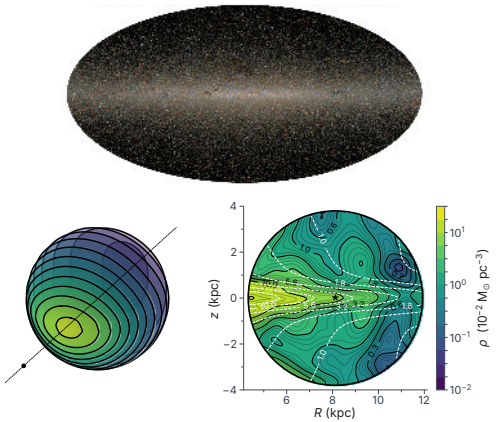
- Fit dark disk + NFW to the 3D $\rho_{\rm DM}$ field at $|z| < 500$ pc
- $10\times$ tighter limit on $\Sigma_{\rm DD}$ at $h_{\rm DD} = 20$ pc, $> 2\times$ at 50–100 pc



95% CL exclusion (pink) vs. Schutz+18, Widmark+21. *Putney et al. 2025*

To recap

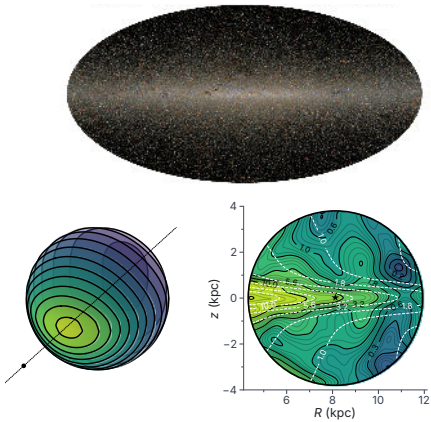
- Flows: the 6D density of 5.8 M *Gaia* stars
- PINN: Boltzmann equation $\rightarrow \Phi$ and dust ϵ, jointly
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$; oblate tilted halo; dark-disk limit



The dust-corrected sky, the potential, and the mass density, all from one fit.

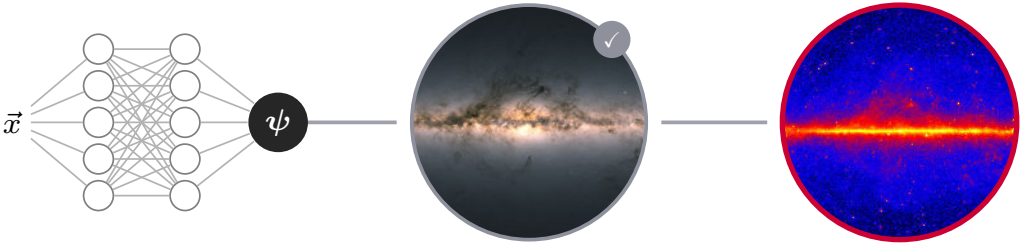
To recap

- Flows: the 6D density of 5.8 M *Gaia* stars
- PINN: Boltzmann equation $\rightarrow \Phi$ and dust ϵ, jointly
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$; oblate tilted halo; dark-disk limit
- Next: *Gaia* DR4 (>100 M 6D stars)
- Approaching the Galactic Center!



The dust-corrected sky, the potential, and the mass density, all from one fit.

Roadmap



Physics-informed neural networks

The PDE residual as the loss function

Part I: ClearPotential

Dark matter from the kinematics of 5.8 M *Gaia* stars

with M. Buckley, S. H. Lim, D. Shih

Part II: PINNCR

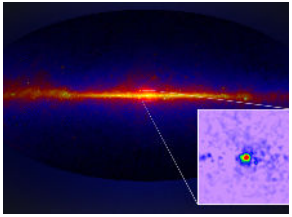
Diffuse γ-ray background to the GeV excess

with Y. Park, D. Shih, M. Buckley, T. Slatyer, C. Chen

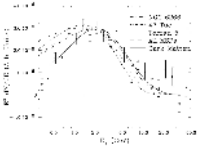
Images: ESA/Gaia/DPAC; NASA/DOE/Fermi-LAT Collaboration

The Galactic Center GeV excess

- GeV γ-ray excess from the inner Milky Way [Goodenough & Hooper 09]



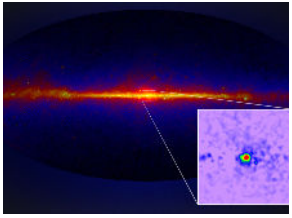
Residual after the diffuse model. *Goodenough & Hooper (2009)*



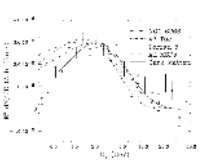
The excess spectrum peaks near 2 GeV. *Daylan et al. (2014)*

The Galactic Center GeV excess

- GeV γ-ray excess from the inner Milky Way [Goodenough & Hooper 09]
- Fits WIMP annihilation, $m_\chi \sim 30$ GeV [Daylan+14]
- Or $\gtrsim 10^4$ unresolved millisecond pulsars [Hooper & Mohlabeng 16]



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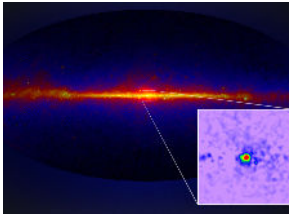




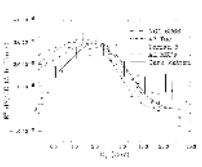
The excess spectrum peaks near 2 GeV. *Daylan et al. (2014)*; MSP art: ESA & F. Ferraro

The Galactic Center GeV excess

- GeV γ-ray excess from the inner Milky Way [Goodenough & Hooper 09]
- Fits WIMP annihilation, $m_\chi \sim 30$ GeV [Daylan+14]
- Or $\gtrsim 10^4$ unresolved millisecond pulsars [Hooper & Mohlabeng 16]
- Point sources or smooth? Tests depend on the background model



Residual after the diffuse model. *Goodenough & Hooper (2009)*





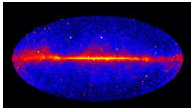
The excess spectrum peaks near 2 GeV. *Daylan et al. (2014)*; MSP art: ESA & F. Ferraro

Where the background comes from

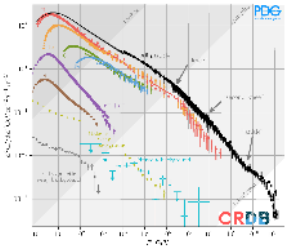
Diffuse γ-rays = cosmic rays (CRs) hitting interstellar gas and light:

- π^0 decay ($p \times$ gas)
- Inverse Compton ($e^- \times$ starlight, CMB)
- Bremsstrahlung ($e^- \times$ gas)

So we need CRs across the *whole* Galaxy, not just what we measure locally.



Fermi-LAT sky, > 1 GeV: mostly diffuse emission.
NASA / DOE / Fermi-LAT Collaboration



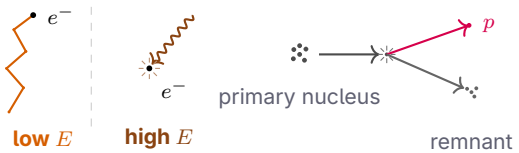
CR spectrum at Earth: one point in the Galaxy. *PDG, Review of Particle Physics 2025*

The cosmic-ray transport equation

The CR density $\psi_i(\vec{r}, p)$ of each species obeys the transport equation:

$$0 = q(\vec{r}, p) + \vec{\nabla} \cdot \left(D_{xx} \vec{\nabla} \psi - \vec{V} \psi \right) + \frac{\partial}{\partial p} p^2 D_{pp} \frac{\partial}{\partial p} \frac{1}{p^2} \psi - \frac{\partial}{\partial p} \left[\dot{p} \psi - \frac{p}{3} (\vec{\nabla} \cdot \vec{V}) \psi \right] - \frac{\psi}{\tau_f} - \frac{\psi}{\tau_r}$$

- Source, diffusion, convection, reacceleration, energy losses, fragmentation, decay.
- Species are coupled.
- Dozens of transport parameters $\vec{\eta}$ (D_0, δ, z_L, injection, ...).

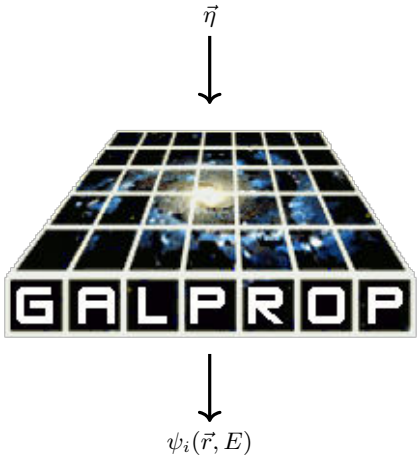


Diffusion, radiative loss, fragmentation.

GALPROP

Standard numerical solver code:

- Pick $\vec{\eta}$, I.C., evolve ψ_i to steady state on a grid.
- CPU cost $\propto N_x^3\, N_E\, N_{\rm species}\, N_{\rm steps}$.
- High res. ≈ 100 CPU-h, memory intensive – *per* $\vec{\eta}$.
- A 10^4-run scan: $\gtrsim$15 CPU-yr, even in 2D.



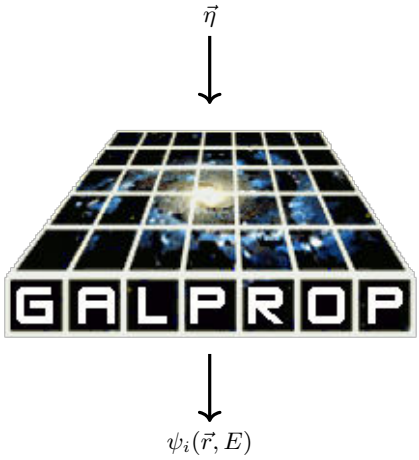
[Strong & Moskalenko 98; Porter+22]

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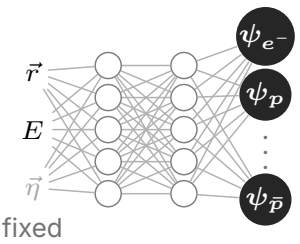
Need a surrogate that returns ψ_i in seconds.

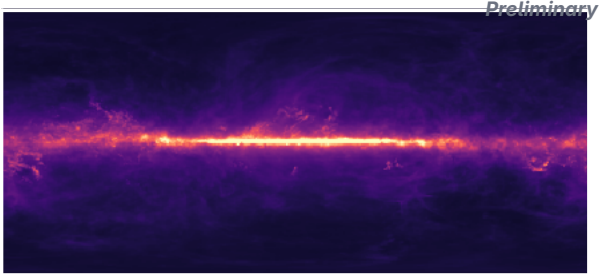


[Strong & Moskalenko 98; Porter+22]

PINNs for cosmic rays and diffuse emission

1. CR flux as a PINN: $\vec{\psi}_\theta(\vec{x}, E)$. Same recipe as Part I: the PDE residual is the loss.

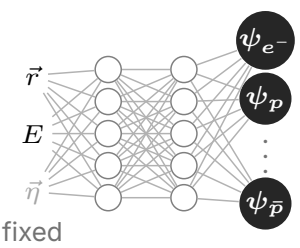


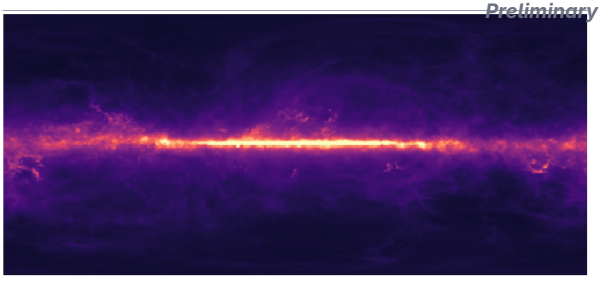


PINN γ-ray sky at 5 GeV: one model (C), then six (A–F).

PINNs for cosmic rays and diffuse emission

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2. Add $\vec{\eta}$ as an input: $\vec{\psi}_\theta(\vec{x}, E, \vec{\eta})$. Fixed $\vec{\eta}$: one GALPROP run.

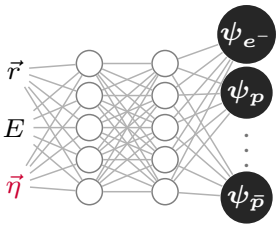


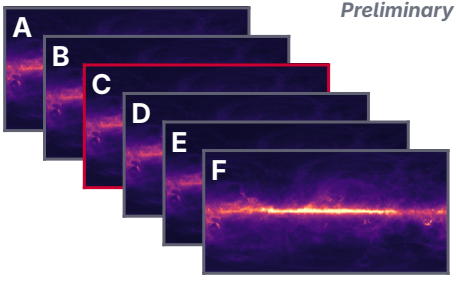


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2. Add $\vec{\eta}$ as an input: $\vec{\psi}_\theta(\vec{x}, E, \vec{\eta})$. Fixed $\vec{\eta}$: one GALPROP run. Varying $\vec{\eta}$: the whole suite.



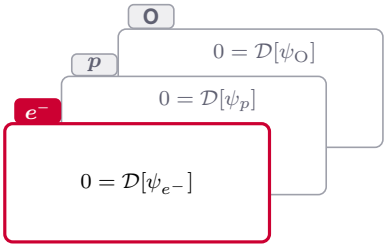


PINN γ-ray sky at 5 GeV: one model (C), then six (A–F).

PINNCR

Differentiable Python code that assembles the transport equation and trains a PINN to solve it:

- **Input:** CR species $\{\psi_i\}$ with their processes and couplings
- **Output:** one network trained on every species' transport residual



$$\mathcal{L} = \frac{1}{N} \sum_i \sum_s w_s \left| \mathcal{D}_s[\psi](\vec{r}_i, E_i, \vec{\eta}) \right|^2$$



Training PINNCR

Standard and non-standard tricks to get PINNCR to converge:

PDE non-dimensionalization

- fields and coefficients scaled to $\mathcal{O}(1)$
- flux in $\log \psi$: 8 decades become one
- log-ratio residual

Network architecture

- boundary built in: $\psi_\theta = B(\vec{r})\mathcal{N}_\theta$, $B = 0$ at the halo edge
- factorized: $\log \psi = \log \phi(\vec{r}, E) + \log S(E)$

Training algorithm

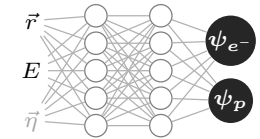
- curriculum: add one process at a time

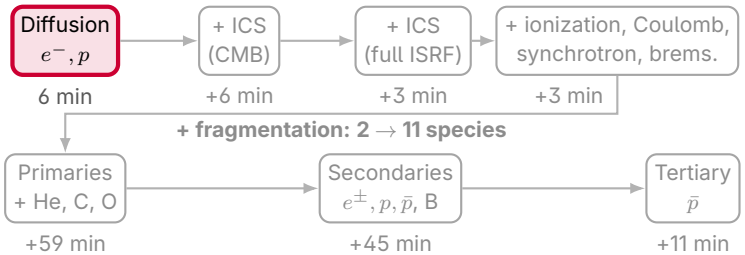
Structured after Wang, Sankaran & Perdikaris, arXiv:2308.08468.

Curriculum learning

Start with simple physics, add processes one at a time.

$$0 = q(\vec{r},p) + \vec{\nabla}\cdot\left(D_{xx}\vec{\nabla}\psi\right) - \vec{\nabla}\cdot(\vec{V}\psi) + \frac{\partial}{\partial p}p^2 D_{pp}\frac{\partial}{\partial p}\frac{1}{p^2}\psi - \frac{\partial}{\partial p}\left[\dot{p}\,\psi - \tfrac{p}{3}(\vec{\nabla}\cdot\vec{V})\psi\right] - \frac{\psi}{\tau_f} - \frac{\psi}{\tau_r}$$



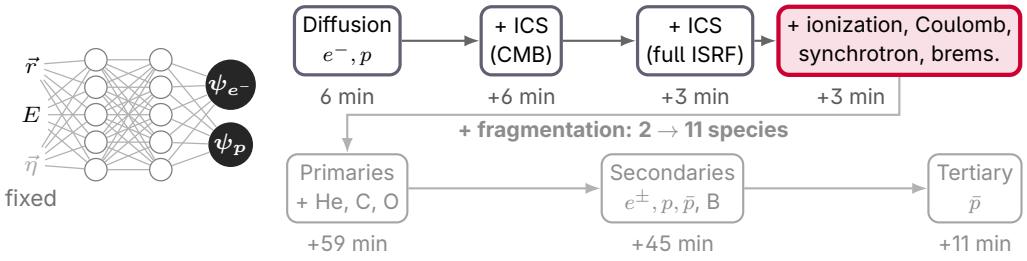


Training time per stage: median over seeds, one GPU.

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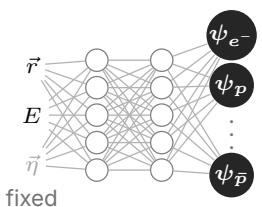


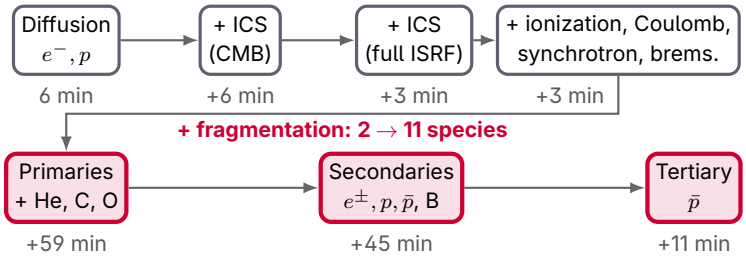
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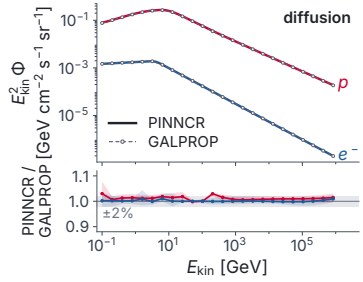
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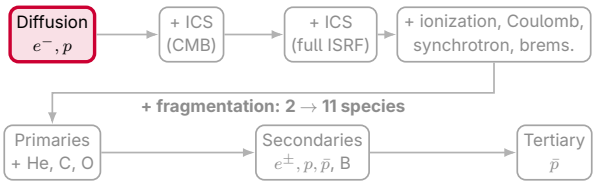
Training time per stage: median over seeds, one GPU.

Diffusion, then energy losses

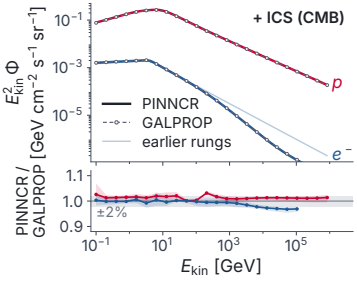


Diffusion + source: e^- and p at the Sun.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.

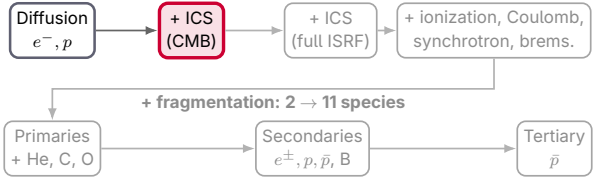


Diffusion, then energy losses

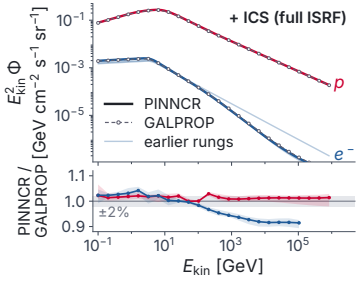


+ inverse Compton on the CMB. Faded: earlier rung.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.

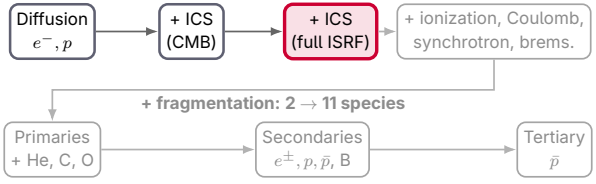


Diffusion, then energy losses

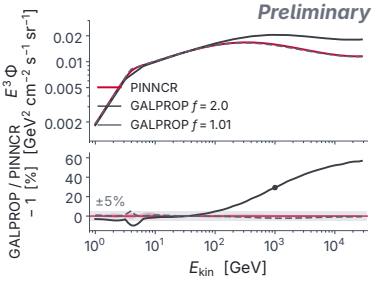


- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.
- + full ISRF: e^- ratio 0.92–0.95 at 1–100 TeV, under study.

+ inverse Compton on the full ISRF (optical + IR).

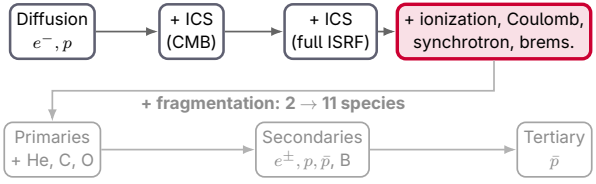


Diffusion, then energy losses

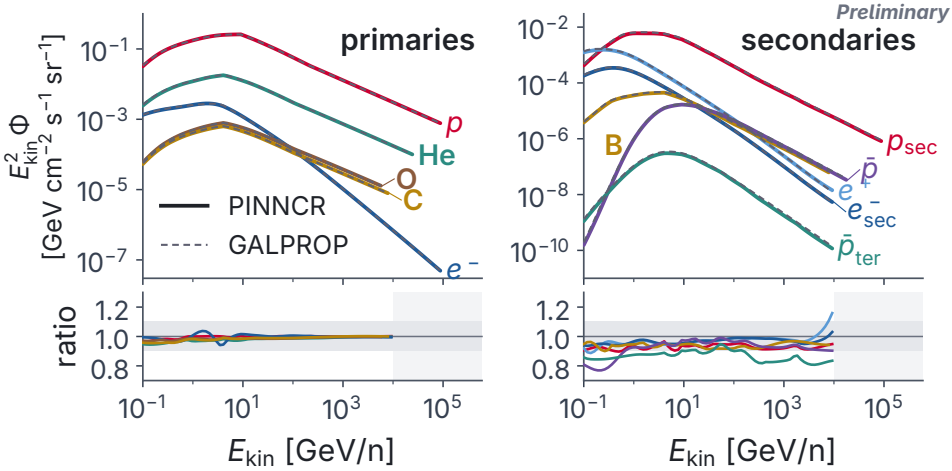


+ ICS and losses: TeV e^- vs. GALPROP's energy grid, $f = E_{i+1}/E_i$.

- Diffusion only: within 1% (e^-) and 3% (p) of GALPROP.
- + ICS on the CMB: e^- within 3% of GALPROP up to 100 TeV.
- + full ISRF: e^- ratio 0.92–0.95 at 1–100 TeV, under study.
- + ICS: GALPROP's default grid is 29% off at 1 TeV; matching PINNCR costs ∼70× the E-bins.

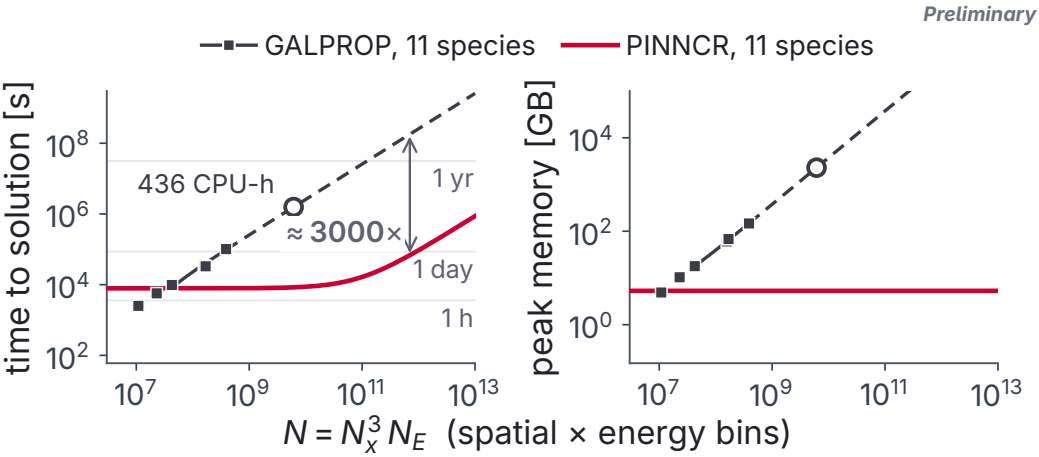


Full-physics GALPROP surrogate model



Same $Z \leq 8$ nuclear network in both codes. Ratio panels: grey band $\pm 10\%$, shaded energies beyond the training range.

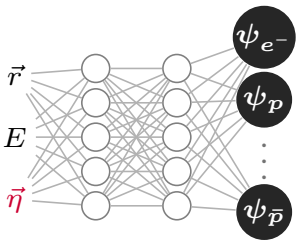
Cost comparison

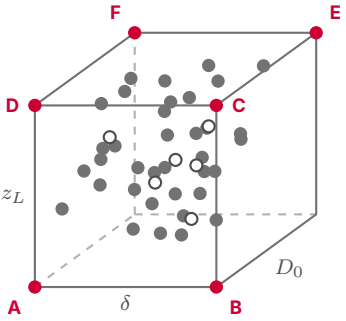


Time to solution and peak memory vs. grid size, 11 species: GALPROP on CPU (squares measured, dashed extrapolated, circle the production run) vs. PINNCR on one GPU.

Varying the transport parameters $\vec{\eta}$

Let $\vec{\eta}$ vary – one network covers a hypercube of diffuse models.



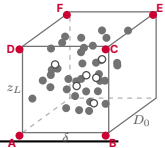


$\vec{\eta}$: 15 parameters (3 axes shown)

- training points (uniform)
- Cholis+21 models A–F
- held-out tests

Fiducial diffuse models

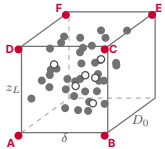
Six published GALPROP models set the training box, each axis padded by 8%.



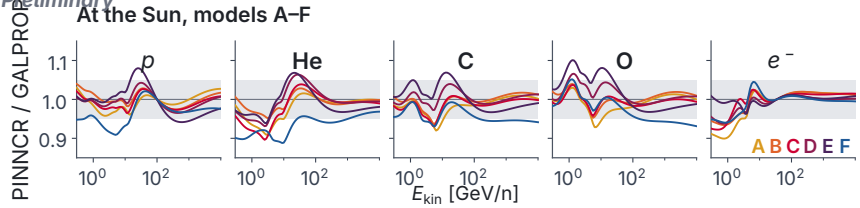
Model	δ	z_L [kpc]	D_0 [10^{28} cm^2/s]	v_A [km/s]	dv_c/dz [km/s/kpc]	injection, H / He γ_1	$R_{\mathrm{br},1}$ [GV]	γ_2	$R_{\mathrm{br},2}$ [GV]	γ_3
A	0.33	5.7	6.70	30.0	0	1.74/1.70	6.0/7.4	2.04/2.16	14.0/21.5	2.41/2.39
B	0.37	5.5	5.50	30.0	2	1.72/1.74	6.0/8.0	2.00/2.14	12.4/21.0	2.38/2.375
C	0.40	5.6	4.85	24.0	1	1.69/1.65	6.0/6.7	2.00/2.13	12.4/20.0	2.38/2.355
D	0.45	5.7	3.90	24.0	5.5	1.69/1.68	6.0/7.0	1.99/2.12	12.4/18.7	2.355/2.34
E	0.50	6.0	3.10	23.0	9	1.71/1.68	6.0/7.2	2.02/2.14	11.2/17.5	2.38/2.33
F	0.43	3.0	1.85	20.0	2	1.68/1.74	6.0/10.5	2.08/2.09	13.0/21.0	2.41/2.33

Cholis et al. (2021), Table I

Accuracy across models A–F



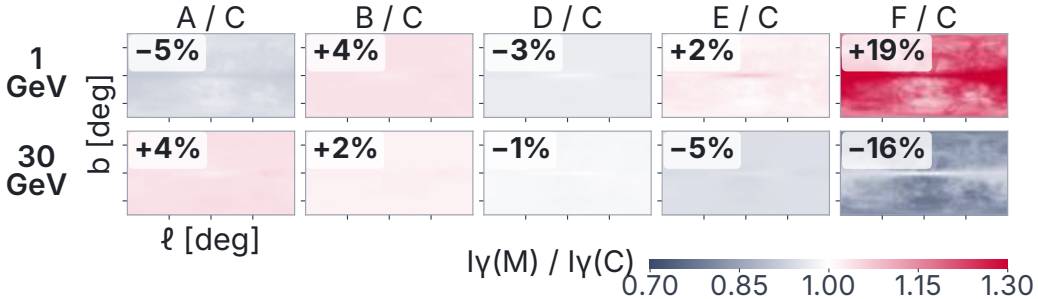
Preliminary



Grey band: ±5%. Medians 1–5% at the Sun (A–F); every disk pixel within 5% at 10 GeV/n (C).

The sky across the fiducial models

Preliminary

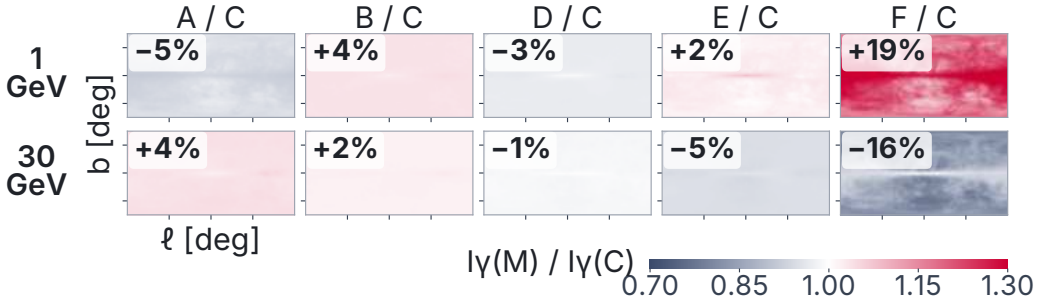


Sky ratio to model C for Cholis+21 models A–F (top 1 GeV, bottom 30 GeV), one PINNCR network.

- Models A–F change the sky by up to 19%.

The sky across the fiducial models

Preliminary

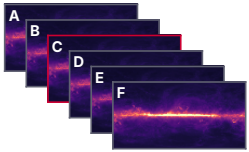


Sky ratio to model C for Cholis+21 models A–F (top 1 GeV, bottom 30 GeV), one PINNCR network.

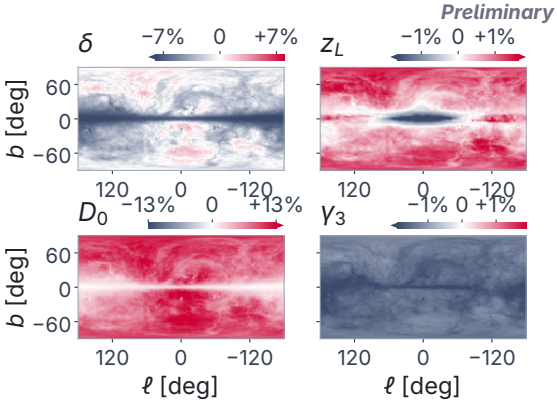
- Models A–F change the sky by up to 19%.
- Now: bracket the excess with six models. Next: a *continuous* posterior over transport.

Differentiable γ-ray maps

- Integrate ψ against gas + ISRF: a γ-ray sky in minutes per $\vec{\eta}$.
- Derivatives $\partial(\text{sky})/\partial\vec{\eta}$: one backward pass.
- Likelihood gradients in $\vec{\eta}$ cost one backward pass: which transport directions mimic the excess?



PINN γ sky, 5 GeV, models A–F.

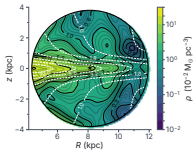


% change of I_γ per transport parameter, model C, 30 GeV.

Summary

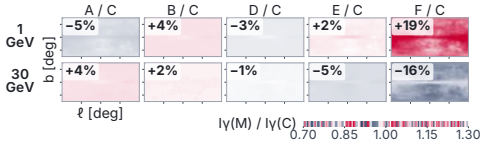
✓ Part I: ClearPotential

- Flows + a PINN: f, Φ and ϵ from 5.8 M *Gaia* stars
- $\bar{\rho}_{\rm DM}(r_\odot) = 0.32 \pm 0.03\,{\rm GeV/cm^3}$
- Oblate tilted halo, dark-disk limit



✓ Part II: PINNCR

- CR transport, 11 species, GALPROP accuracy, no grid
- One network for different η; γ sky + gradients in minutes
- Next: the transport-marginalized GeV excess fit



Thanks for listening!

`arXiv:2412.14236`, `2512.09989` (ClearPotential) · `arXiv:2610.XXXXX` + GitHub (PINNCR)